Programming Fundamental lab "2"

Task 1:

Type and save the following programs in visual studio. Run these programs and observe their output.

Input I

```
#include<iostream>
 Using namespace std;
 int main()
{
 char letter;
 letter = 'A';
 cout << letter << '\n';
  letter = 'B';
 cout << letter << '\n';
 cin >> letter;
 return 0;
}
```

Output I

```
A
B
```

Input II

```
#include<iostream>
 using namespace std;
 int main()
{
 char letter;
 letter = 65;
 cout << letter << '\n';
 letter = 66;
 cout << letter << '\n';
 cin >> letter;
  return 0;
}
```

Output II

```
A
B
```

Input III

```
#include<iostream>
using namespace std;
int main()
{
 int letter;
 letter = 'A';
 cout << letter << '\n';
 letter = 'B';
 cout << letter << '\n';
```

Output III

```
65
66
```

Task 2:

Take two integers as input from the user apply arithematic operations on them (+,-,*,/) as print them on screen.

Input:

```
#include<iostream>
using namespace std;
 int main()
 {
  int a,b;
  cout<<"Enter First number:";
  cin>>a;
   cout<<"Enter second number: ";
   cin>>b;
   cout<<"Sum="<<a+b<<endl;
   cout<<"Substraction="<<a-b<<endl;
   cout<<"Multiplication="<<a*b<<endl;
   if (b!=0)
      cout<<"Division="<<a/b<<endl;
      else
        cout<<"Division=Not possible(divided by zero)"<<endl;
        return 0;

    }
```

Output:

```
Enter first number: 10
Enter second number:5

Sum= 15
Subtraction=5
Multiplication = 50
Division=2
```

Task 3:

Write the output of the following code.

```
#include<iostream>
 Using namespace std;
 int main()
 {
   int val1;
    int val2;
    int val3;
    cout<<"Please enter your 3 numbers:";
    cin>>val1>> val2>>val3;
    cout<<val1<<"\n"<<val2<<"\n"<<val3<<"\n";
    cout<<val3<<"\n"<<val2<<"\n"<<val1<<"\n";
     return 0;
    }
```

Output:

```
10
20
30
30
20
10
```

Task 4:

Write a C++ program to calculate the distance between two points. Note x1, y1, y2 are all double values.

Formula:

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Input code:

```
#include<iostream>
using namespace std;
int main(){
    double x1, y1, x2, y2;
    cout<<"Enter x2 y3 x5 y7 :";
    cin>>x1>>y1>>x2>>y2;
    double d = pow(x2-x1,2)+Pow(y2-y1,2);
    cout<<"Distance="<<d;
    return 0;
}
```

Output:

Enter x2 y3 x5 y7

Distance = 25

Task 5:

i) Calculate the area of a circle (area, PI*r2)

Input code:

```
#include<iostream>
using namespace std;
int main(){
    double length, width, area;
    cout<<"Enter length and width: 4 6";
    cin>>length>>width;
    area = length * width;
    cout<<"Area of Rectangle=24"<< area;
```

```
return 0;
}
```

Output:

```
Enter length and width 4 6
Area of Rectangle = 24
```

ii) Calculate the area of a Circle (area = Pi * r2)

```
#include<iostream>
using namespace std;
int main(){
 double r = 5;
 double Pi = 3.1416;
 double area = Pi * r * r;
 cout<< "Enter radius : "<< r << endl;
 cout<< "Area of circle = "<< area;
  return 0;
}
```

Output:

```
Enter radius: 5
Area of Circle = 78.54
```

iii) Calculate the Area of a Triangle (area = base * height * .5)

```
#include<iostream>
  using namespace std;
  int main(){
    double base = 4, height = 5;
    double area = base * height * 0.5;
    cout<< "Enter base and height: "<<base<< " "<<height<<endl;
    cout<< "Area of Triangle = "<<area;
    return 0;

  }
```

Output:

```
Enter base and height: 4 5
Area of Triangle = 10
```

Task 6:

Total Purchase A customer in a store is purchasing five items. The prices of the five items are:

Price of item 1 = $ 12.95
Price of item 2 = $ 24.95
Price of item 3 = $ 6.95
Price of item 4 = $ 14.95
Price of item 5 = $ 3.95

Input:

```
#include<iostream>
 using namespace std;
 int main(){
     double item = 12.95
     double item = 24.95
     double item = 6.95
      double item = 14.95
      double item = 3.95
     double subtotal = item1+item2+item3+item4+item5;
     double salesTax = subtotal * 0.06; //6% tax
     double total = subtotal + salesTax;
     cout<<"Price of item 1 = $" <<endl;
     cout<<" Price of item 2 = $" <<endl;
     cout<<" Price of item 3 = $" <<endl;
     cout<<"Price of item 4 = $" <<endl;
     cout<<" Price of item 5 = $" <<endl;
     cout<<endl;
     cout<<" Subtotal = $" << subtotal <<endl;
     cout <<" sales Tax (6%) = $<< sales tax <<endl;
      cout <<" Total = $" <<endl;
      return 0;

     }
```

Output:

Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95

Subtotal = $63.75
Sales Tax (6%) = $3.825
Total = $67.575